

## 4.B The Hopf Invariant

In §2.2 we used homology to distinguish different homotopy classes of maps  $S^n \rightarrow S^n$  via the notion of degree. We will show here that cup product can be used to do something similar for maps  $S^{2n-1} \rightarrow S^n$ . Originally this was done by Hopf using more geometric constructions, before the invention of cohomology and cup products.

In general, given a map  $f: S^m \rightarrow S^n$  with  $m \geq n$ , we can form a CW complex  $C_f$  by attaching a cell  $e^{m+1}$  to  $S^n$  via  $f$ . The homotopy type of  $C_f$  depends only on the homotopy class of  $f$ , by Proposition 0.18. Thus for maps  $f, g: S^m \rightarrow S^n$ , any invariant of homotopy type that distinguishes  $C_f$  from  $C_g$  will show that  $f$  is not homotopic to  $g$ . For example, if  $m = n$  and  $f$  has degree  $d$ , then from the cellular chain complex of  $C_f$  we see that  $H_n(C_f) \approx \mathbb{Z}_{|d|}$ , so the homology of  $C_f$  detects the degree of  $f$ , up to sign. When  $m > n$ , however, the homology of  $C_f$  consists of  $\mathbb{Z}$ 's in dimensions 0,  $n$ , and  $m + 1$ , independent of  $f$ . The same is true of cohomology groups, but cup products have a chance of being nontrivial in  $H^*(C_f)$  when  $m = 2n - 1$ . In this case, if we choose generators  $\alpha \in H^n(C_f)$  and  $\beta \in H^{2n}(C_f)$ , then the multiplicative structure of  $H^*(C_f)$  is determined by a relation  $\alpha^2 = H(f)\beta$  for an integer  $H(f)$  called the **Hopf invariant** of  $f$ . The sign of  $H(f)$  depends on the choice of the generator  $\beta$ , but this can be specified by requiring  $\beta$  to correspond to a fixed generator of  $H^{2n}(D^{2n}, \partial D^{2n})$  under the map  $H^{2n}(C_f) \approx H^{2n}(C_f, S^n) \rightarrow H^{2n}(D^{2n}, \partial D^{2n})$  induced by the characteristic map of the cell  $e^{2n}$ , which is determined by  $f$ . We can then change the sign of  $H(f)$  by composing  $f$  with a reflection of  $S^{2n-1}$ , of degree  $-1$ . If  $f \simeq g$ , then under the homotopy equivalence  $C_f \simeq C_g$  the chosen generators  $\beta$  for  $H^{2n}(C_f)$  and  $H^{2n}(C_g)$  correspond, so  $H(f)$  depends only on the homotopy class of  $f$ .

If  $f$  is a constant map then  $C_f = S^n \vee S^{2n}$  and  $H(f) = 0$  since  $C_f$  retracts onto  $S^n$ . Also,  $H(f)$  is always zero for odd  $n$  since in this case  $\alpha^2 = -\alpha^2$  by the commutativity property of cup product, hence  $\alpha^2 = 0$ .

Three basic examples of maps with nonzero Hopf invariant are the maps defining the three Hopf bundles in Examples 4.45, 4.46, and 4.47. The first of these Hopf maps is the attaching map  $f: S^3 \rightarrow S^2$  for the 4-cell of  $CP^2$ . This has  $H(f) = 1$  since  $H^*(\mathbb{C}P^2; \mathbb{Z}) \approx \mathbb{Z}[\alpha]/(\alpha^3)$  by Theorem 3.12. Similarly,  $\mathbb{H}P^2$  gives rise to a map  $S^7 \rightarrow S^4$  of Hopf invariant 1. In the case of the octonionic projective plane  $\mathbb{O}P^2$ , which is built from the map  $S^{15} \rightarrow S^8$  defined in Example 4.47, we can deduce that  $H^*(\mathbb{O}P^2; \mathbb{Z}) \approx \mathbb{Z}[\alpha]/(\alpha^3)$  either from Poincaré duality as in Example 3.40 or from Exercise 5 for §4.D.

It is a fundamental theorem of [Adams 1960] that a map  $f: S^{2n-1} \rightarrow S^n$  of Hopf invariant 1 exists only when  $n = 2, 4, 8$ . This has a number of very interesting consequences, for example:

- $\mathbb{R}^n$  is a division algebra only for  $n = 1, 2, 4, 8$ .
- $S^n$  is an H-space only for  $n = 0, 1, 3, 7$ .
- $S^n$  has  $n$  linearly independent tangent vector fields only for  $n = 0, 1, 3, 7$ .
- The only fiber bundles  $S^p \rightarrow S^q \rightarrow S^r$  occur when  $(p, q, r) = (0, 1, 1), (1, 3, 2), (3, 7, 4)$ , and  $(7, 15, 8)$ .

The first and third assertions were in fact proved shortly before Adams' theorem in [Kervaire 1958] and [Milnor 1958] as applications of a theorem of Bott that  $\pi_{2n}U(n) \approx \mathbb{Z}_n$ . A full discussion of all this, and a proof of Adams' theorem, is given in [VBKT].

Though maps of Hopf invariant 1 are rare, there are maps  $S^{2n-1} \rightarrow S^n$  of Hopf invariant 2 for all even  $n$ . Namely, consider the space  $J_2(S^n)$  constructed in §3.2. This has a CW structure with three cells, of dimensions 0,  $n$ , and  $2n$ , so  $J_2(S^n)$  has the form  $C_f$  for some  $f: S^{2n-1} \rightarrow S^n$ . We showed that if  $n$  is even, the square of a generator of  $H^n(J_2(S^n); \mathbb{Z})$  is twice a generator of  $H^{2n}(J_2(S^n); \mathbb{Z})$ , so  $H(f) = \pm 2$ .

From this example we can get maps of any even Hopf invariant when  $n$  is even via the following fact.

**Proposition 4B.1.** *The Hopf invariant  $H: \pi_{2n-1}(S^n) \rightarrow \mathbb{Z}$  is a homomorphism.*

**Proof:** For  $f, g: S^{2n-1} \rightarrow S^n$ , let us compare  $C_{f+g}$  with the space  $C_{f \vee g}$  obtained from  $S^n$  by attaching two  $2n$ -cells via  $f$  and  $g$ . There is a natural quotient map  $q: C_{f+g} \rightarrow C_{f \vee g}$  collapsing the equatorial disk of the  $2n$ -cell of  $C_{f+g}$  to a point. The induced cellular chain map  $q_*$  sends  $e_{f+g}^{2n}$  to  $e_f^{2n} + e_g^{2n}$ . In cohomology this implies that  $q^*(\beta_f) = q^*(\beta_g) = \beta_{f+g}$  where  $\beta_f, \beta_g$ , and  $\beta_{f+g}$  are the cohomology classes dual to the  $2n$ -cells. Letting  $\alpha_{f+g}$  and  $\alpha_{f \vee g}$  be the cohomology classes corresponding to the  $n$ -cells, we have  $q^*(\alpha_{f \vee g}) = \alpha_{f+g}$  since  $q$  is a homeomorphism on the  $n$ -cells. By restricting to the subspaces  $C_f$  and  $C_g$  of  $C_{f \vee g}$  we see that  $\alpha_{f \vee g}^2 = H(f)\beta_f + H(g)\beta_g$ . Thus  $\alpha_{f+g}^2 = q^*(\alpha_{f \vee g}^2) = H(f)q^*(\beta_f) + H(g)q^*(\beta_g) = (H(f) + H(g))\beta_{f+g}$ .  $\square$

**Corollary 4B.2.**  *$\pi_{2n-1}(S^n)$  contains a  $\mathbb{Z}$  direct summand when  $n$  is even.*

**Proof:** Either  $H$  or  $H/2$  is a surjective homomorphism  $\pi_{2n-1}(S^n) \rightarrow \mathbb{Z}$ .  $\square$

## Exercises

1. Show that the Hopf invariant of a composition  $S^{2n-1} \xrightarrow{f} S^{2n-1} \xrightarrow{g} S^n$  is given by  $H(gf) = (\deg f)H(g)$ , and for a composition  $S^{2n-1} \xrightarrow{f} S^n \xrightarrow{g} S^n$  the Hopf invariant satisfies  $H(gf) = (\deg g)^2 H(f)$ .
2. Show that if  $S^k \rightarrow S^m \xrightarrow{p} S^n$  is a fiber bundle, then  $m = 2n - 1$ ,  $k = n - 1$ , and, when  $n > 1$ ,  $H(p) = \pm 1$ . [Show that  $C_p$  is a manifold and apply Poincaré duality.]